Matthew Snelgrove

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matthewSnelgrove
portfolio

Education

University of Toronto, Scarborough | **Computer Science** | CGPA: 3.72 September 2021 – Present
Courses: Data Structures, Algorithms, Software Design, Security, Databases, Operating Systems, Web Development

Experience

SS&C Technologies (IFDS) | **Software Developer** September 2023 – April 2024

- Eliminated 3 hours of managerial work per week by automating component dependency tracking with internal APIs, and visualized output in graphical and tabular views in a modern React web app.
- Ensured the maintainability of the project and expedited its handover by documenting the codebase with jsDoc and creating comprehensive wiki entries.
- Improved system reliability by identifying and patching production bugs across team apps rapidly.
- Accelerated modernization projects by investigating internal tools and delivering actionable user guides to streamline their integration into existing systems.

Ontario Ministry of Transportation | **Junior Technical Analyst** January 2023 – April 2023

- Developed RESTful API for modernization of critical driver/vehicle registration systems with Java Spring Boot.
- Cut debugging time by over 50% through automated unit and integration testing with JUnit and Postman, achieving 100% code coverage and preemptively identifying errors caused pending dependencies changes.
- Recovered ~3 hours per week of developer time by building and deploying new versions with Maven and Azure Pipelines instead of time-consuming and error-prone manual builds.

Projects

Superbiddo October 2024 – December 2024

- Implemented core features for the auction-based marketplace app by developing a flexible and scalable RESTful API using Express and a relational database using PostgreSQL.
- Deployed app with a three-tier architecture on a Google Cloud VM using Docker, enabling targeted optimization, independent updates to each layer, and robust scalability.
- Ensured reliability, security, and ample performance at minimal cost by customizing a Google Cloud VM to host the containerized application, setting up monitoring and alerts, and enforcing secure network configurations.
- Automated consistent and reliable deployments with a bash script to build images, upload them to the VM, and start containers with secrets using a Docker Compose file.
- Fully secured the app and eliminated web vulnerabilities with helmet.js, HTTPS, data validation, CSRF protection, and session-based authentication.

Pintos Operating System May 2024 – August 2024

- Extended skeleton code for a basic operating system in C to support system calls, indexed and extensible files, and ELF program execution while ensuring stability and performance in a multithreaded environment.
- Improved fairness and responsiveness by implementing priority donation and a multi-level feedback queue scheduler (MLFQS) to dynamically adjust thread priorities based on CPU usage and workload.
- Optimized memory management efficiency by designing a virtual memory system with page fault handling, LRU-approximated page replacement with the clock algorithm, and swap space utilization.

Technical Skills

Languages: Java, JavaScript/TypeScript, Python, C/C++, C#, SQL (Postgres), HTML, CSS
Frameworks & Libraries: Node.js, Express.js, Spring Boot, React, MongoDb, JUnit, jQuery
Developer Tools: Git, Docker, Google Cloud Platform, Azure, Postman, OpenApi